



AI AND MENTAL HEALTH:

AN INTRODUCTION

Simone J. Skeen

02.25



AI: _____

AI: the capability of computer systems to perform complex tasks [typically] associated with human intelligence...¹

1. Grand Challenges Canada, McKinsey Health Institute, Google. *Mental Health and AI Field Guide*. July 7, 2025.



AI: the capability of computer systems to perform complex tasks associated with human intelligence...

such as learning, decision-making, and creating.¹

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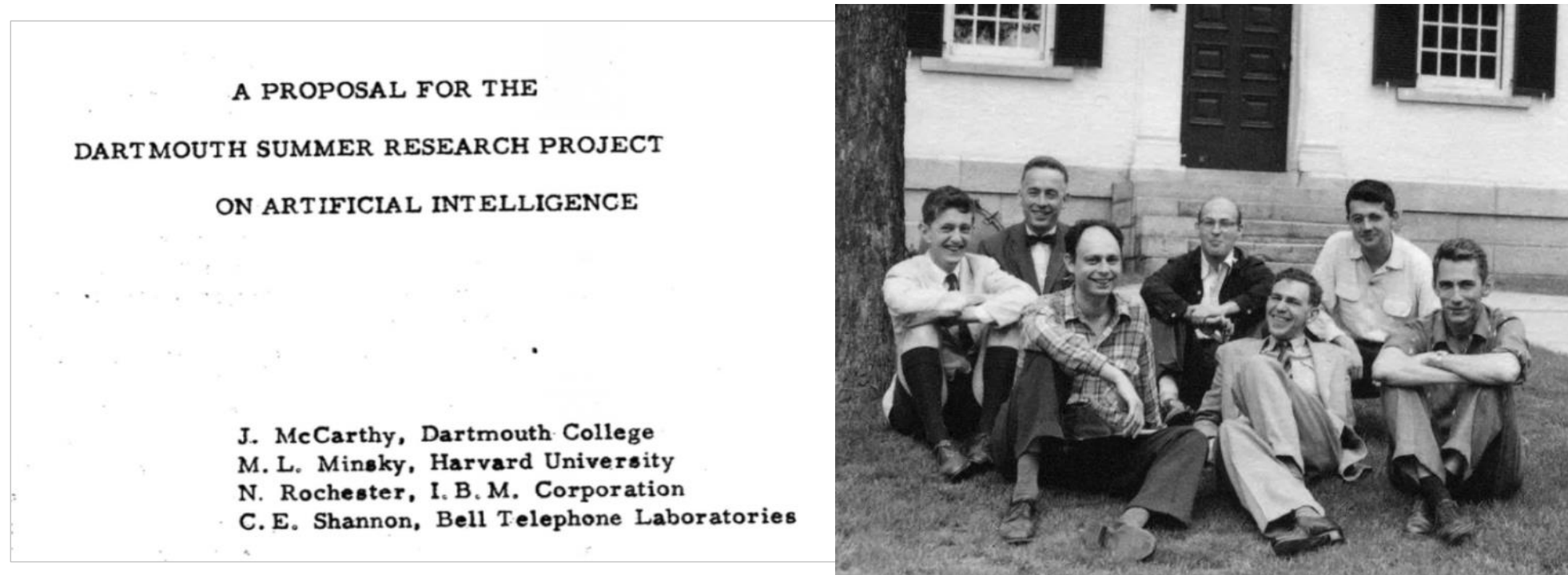


Fig. 1. The 1956 Dartmouth Project: Rockefeller Foundation proposal (*left*); attendees (*right*).²

2. Strickland E. The turbulent past and uncertain future of artificial intelligence. *IEEE Spectrum*. September 30, 2021.

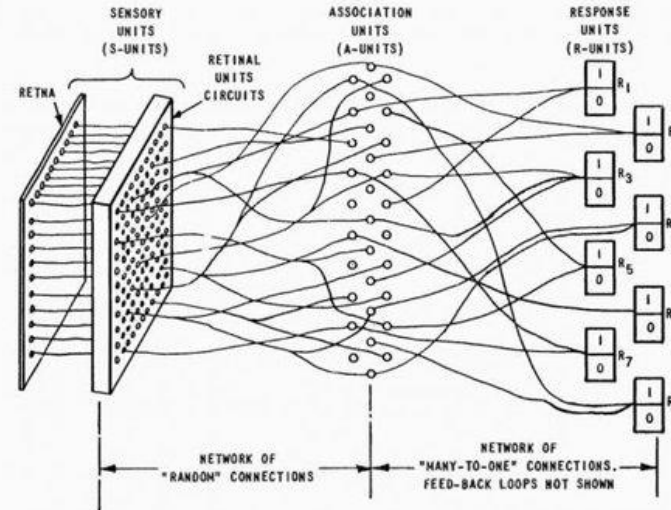
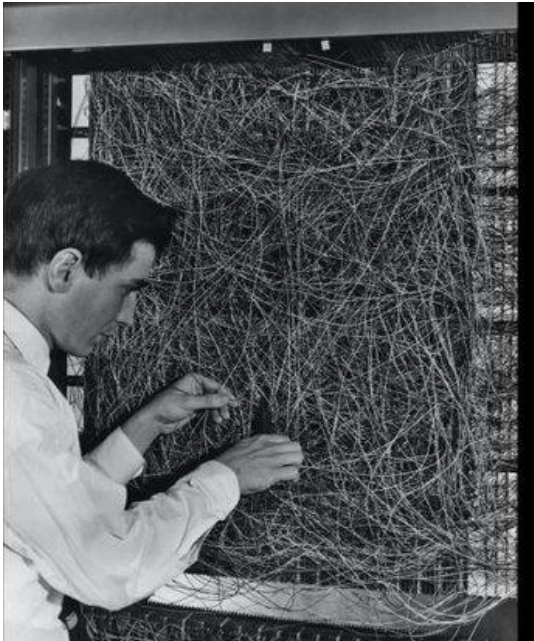


Figure 1 ORGANIZATION OF THE MARK I PERCEPTRON

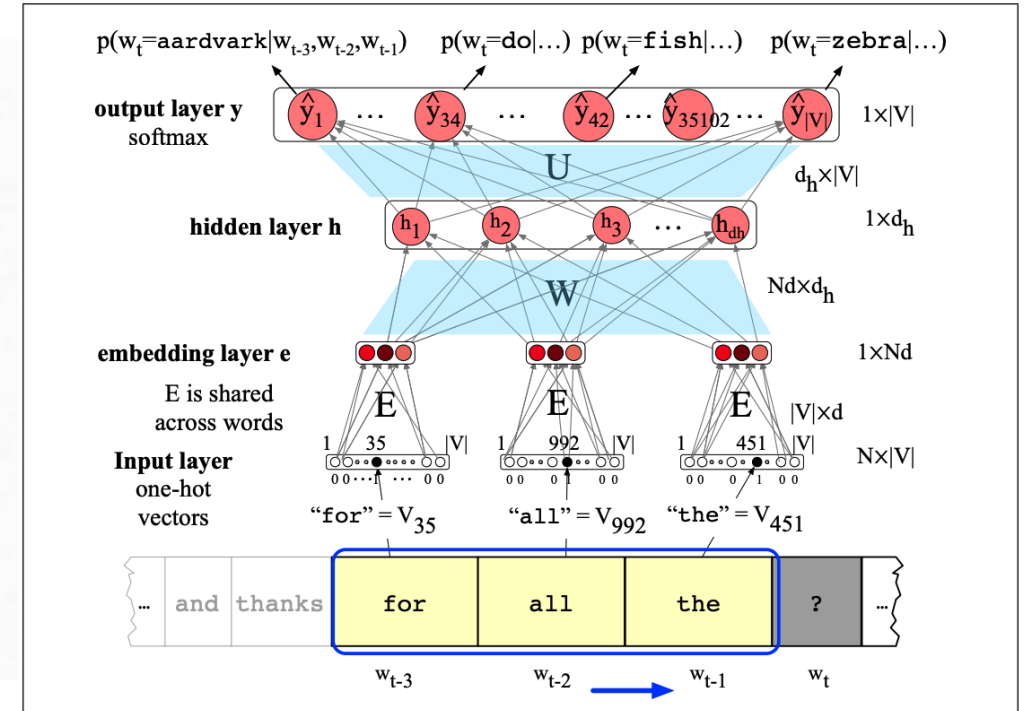


Fig. 2. The Mark I Perceptron, ~1958 (left);³ a feedforward neural language model, 2026 (right).⁴

3. Lefkowitz M. Professor's perceptron paved the way for AI – 60 years too soon. *Cornell Chronicle*. September 25, 2019.

4. Jurafsky D, Martin JH. Neural networks. In: *Speech and Language Processing*. 3rd ed. January 6, 2026.

■ *Deterministic* ↔ ○ *Probabilistic*

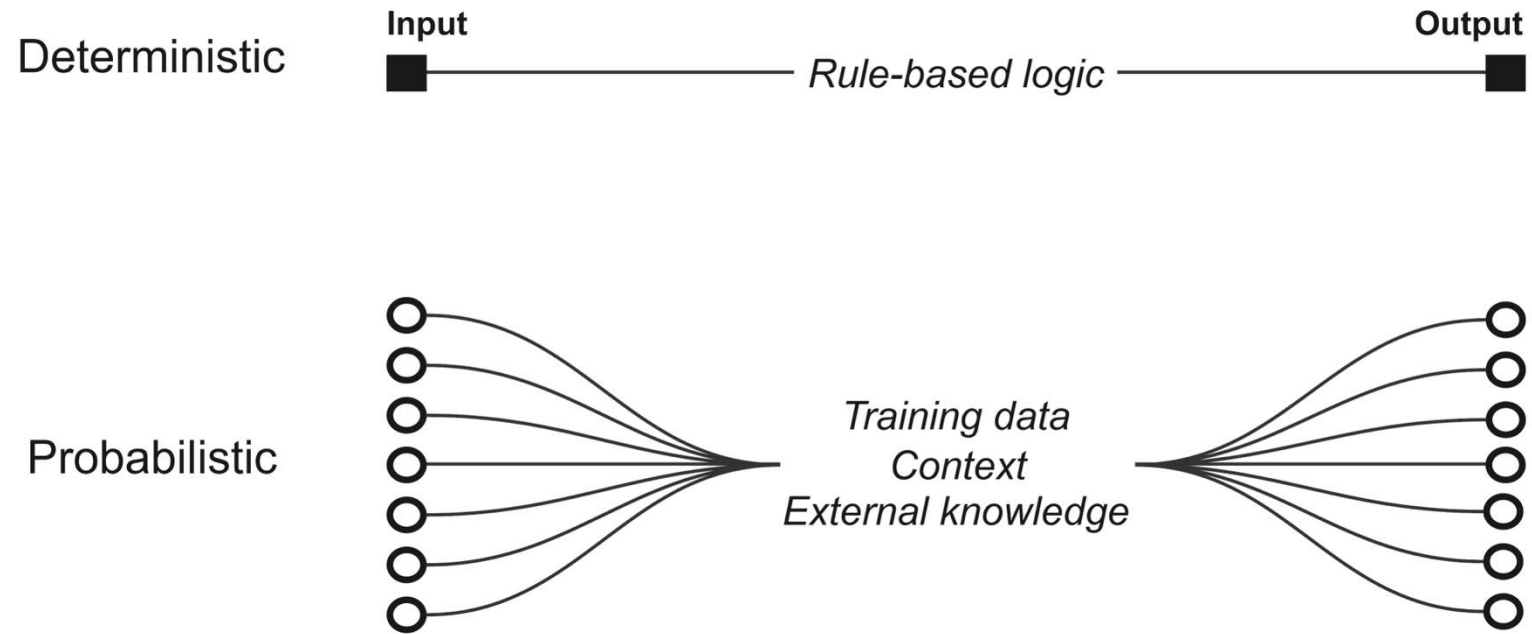


Fig. 3. Deterministic versus probabilistic system design (adapted from Morgan).^{5,6}

5. Morgan B. AI in product design concepts: a primer on deterministic vs probabilistic systems. *Bootcamp*. November 17, 2024.
6. Jurafsky D, Martin JH. Large language models. In: *Speech and Language Processing*. 3rd ed. January 6, 2026.

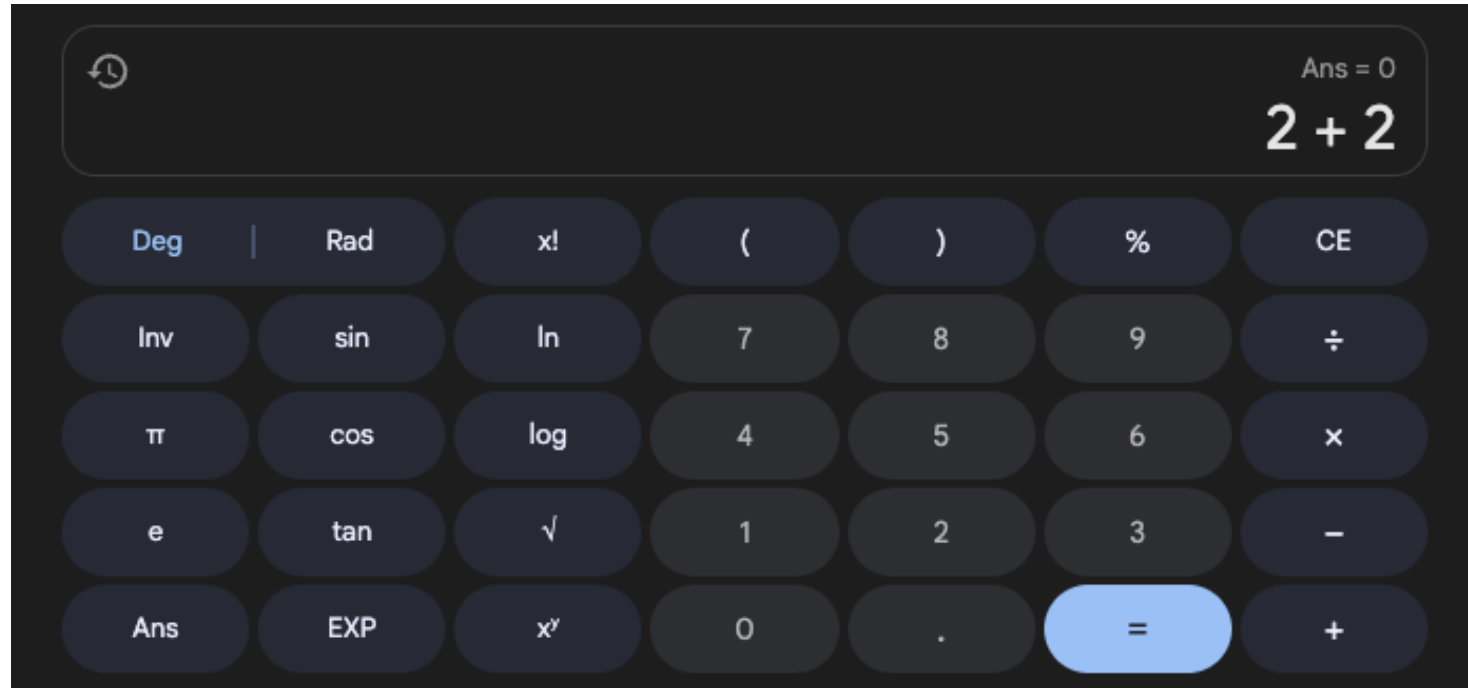


Fig. 4. A calculator (Google).

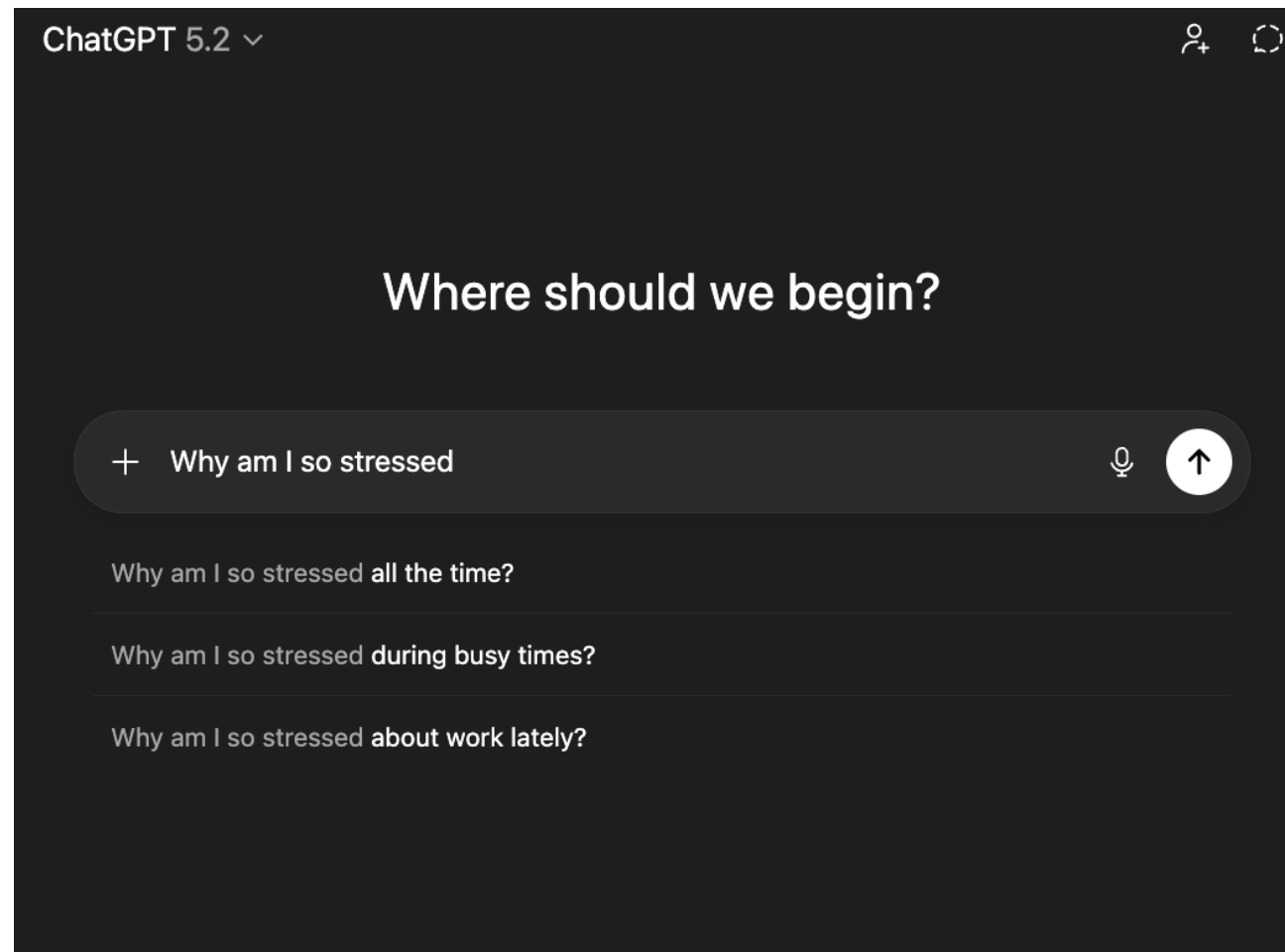


Fig. 5. ChatGPT.⁷

7. OpenAI. ChatGPT 5.2. Accessed February 5, 2026.



Discriminative



Generative

 ***Discriminative:*** learns the boundaries between different classes of data, determines the most accurate *label or decision* for a given input.^{8,9}

8. IBM. What is an AI model? Generative models vs. discriminative models. n.d.

9. Ng AY, Jordan MI. On discriminative vs. generative classifiers: a comparison of logistic regression and naïve Bayes. *Adv Neural Inf Process Syst*. 2001;14:841-848.



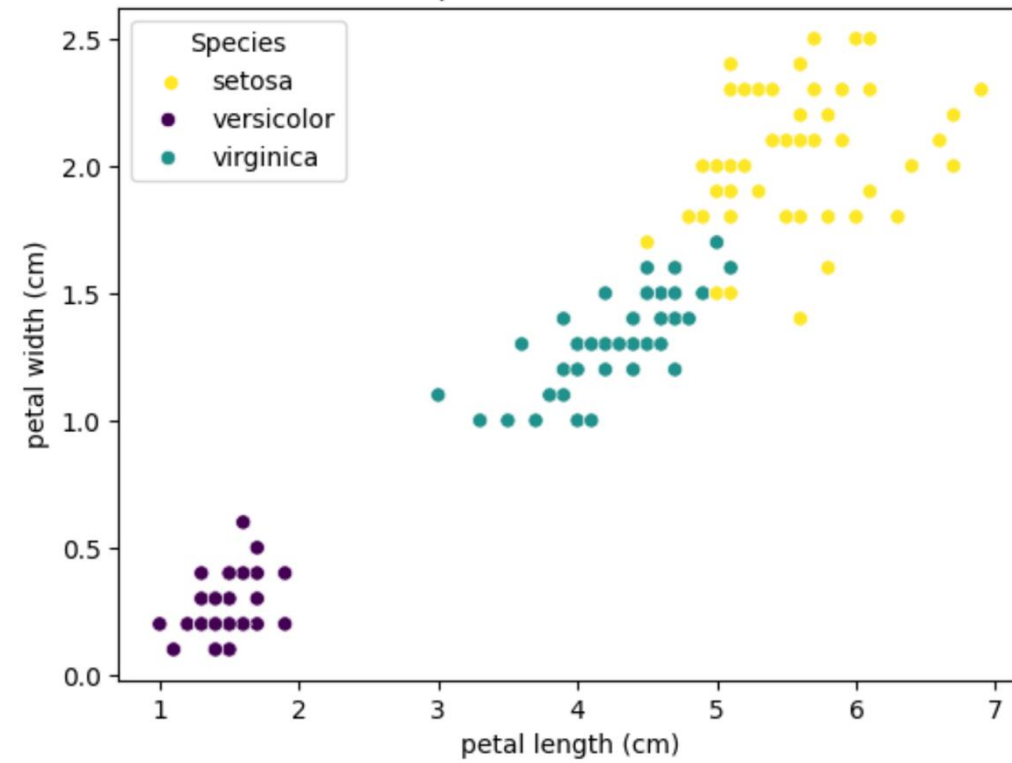


Fig. 6a. Petal measurements of iris species.^{10,11*}

¹⁰. Törnberg P. Machine learning. Presented as part of SICSS: Summer Institutes in Computational Social Science; June 2024; University of Amsterdam, NL.

¹¹. Unwin A, Kleinman K. The iris data set: in search of the source of *virginica*. *Signif.* 2021;18:26-29.

* Adapted from Törnberg.

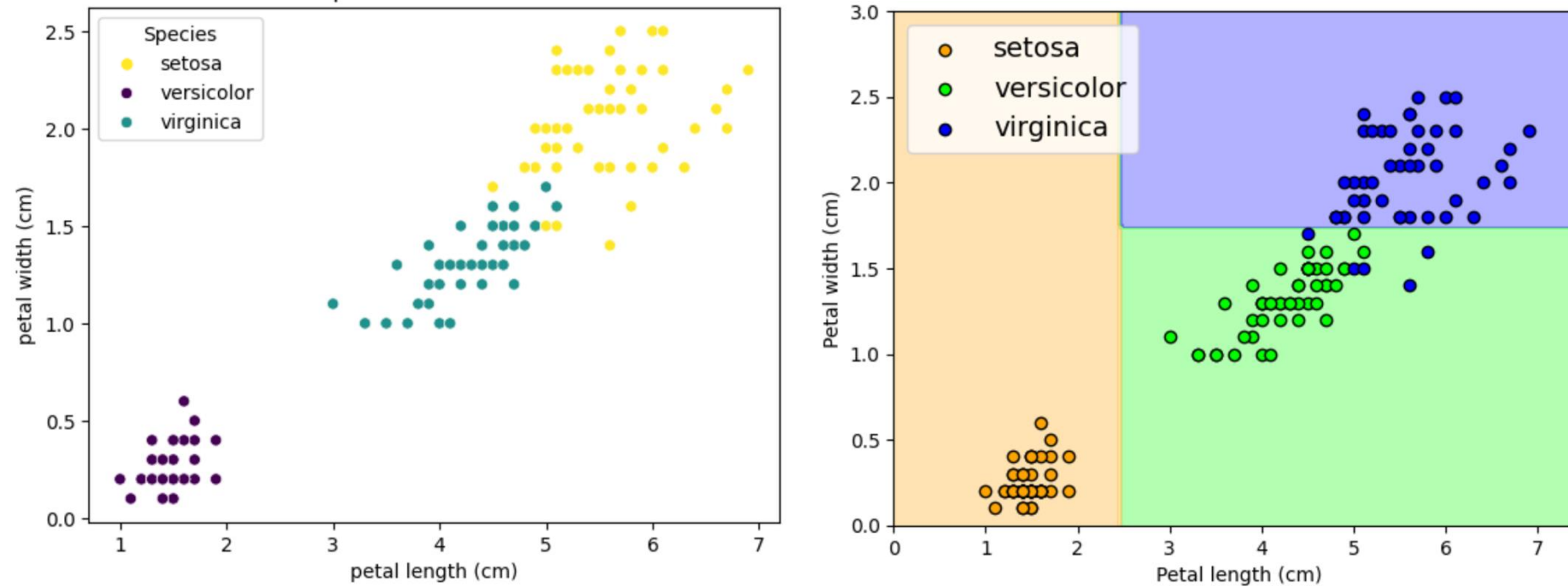


Fig. 6a-b. Petal measurements of iris species (*left*); iris classification model decision boundaries (*right*).^{10,11*}

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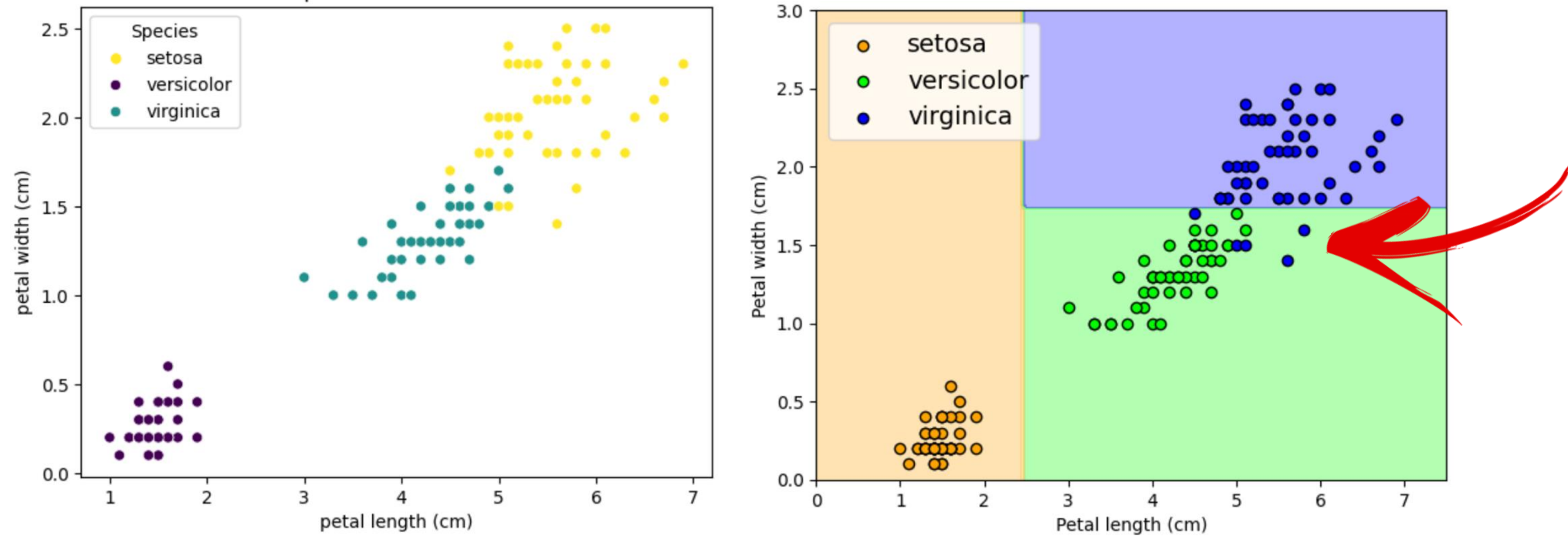


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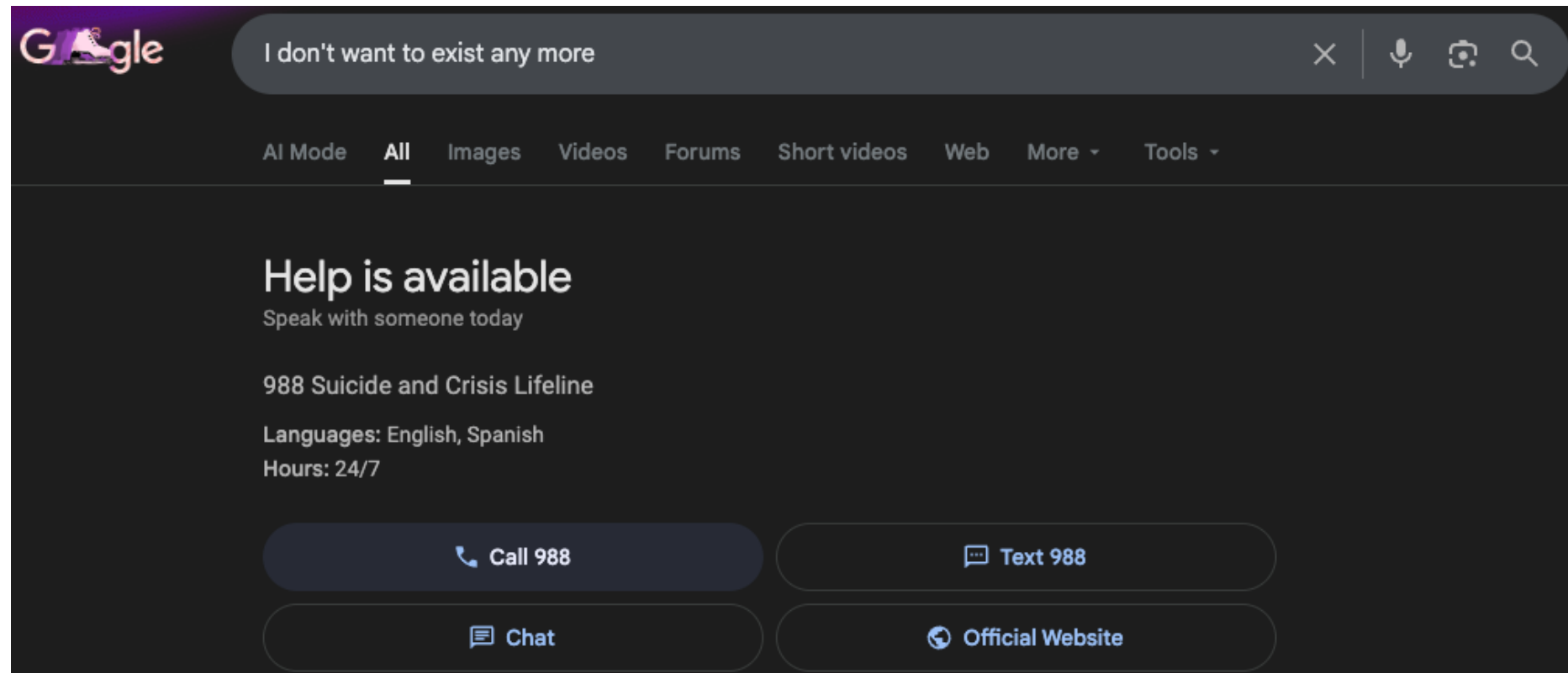


Fig. 7. Search engine retrieval (Google).

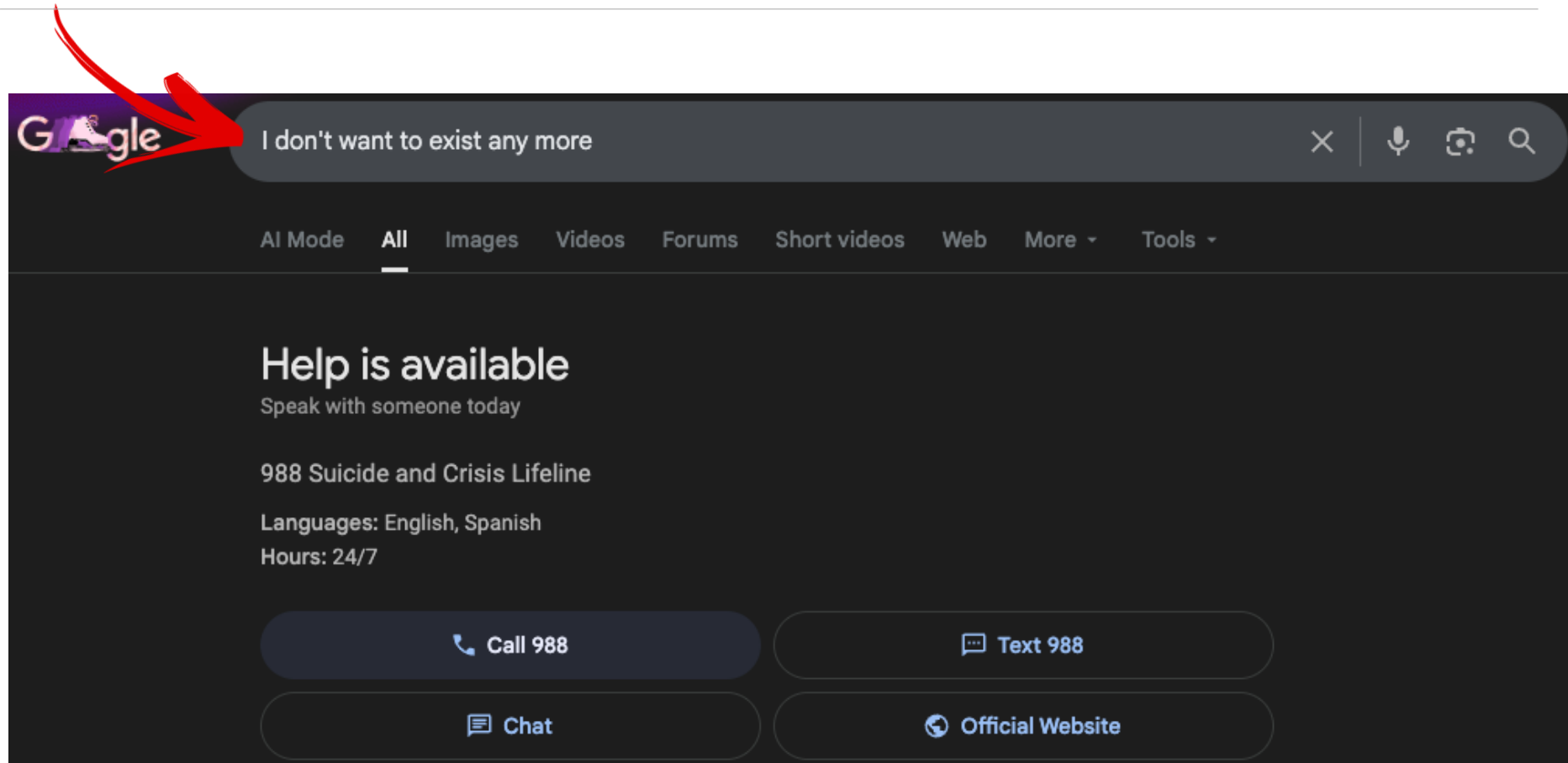


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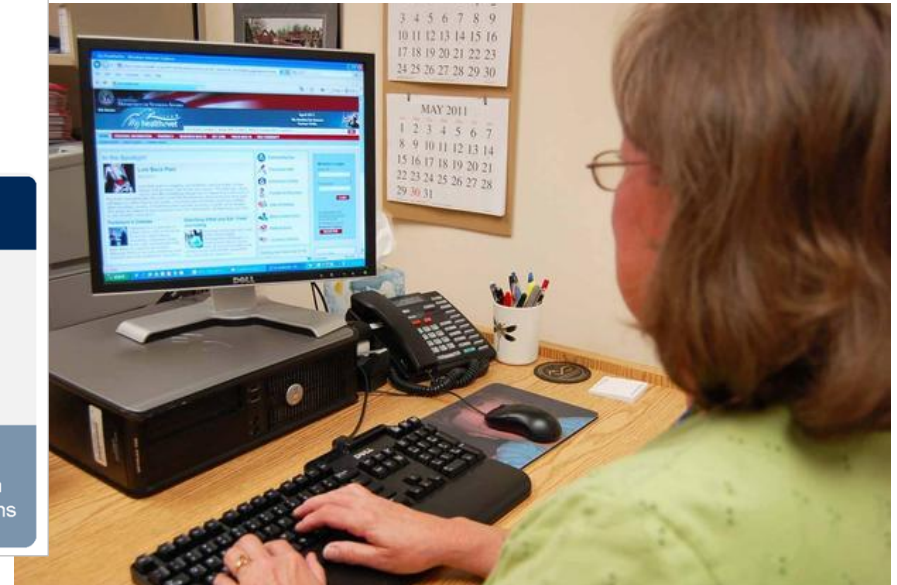
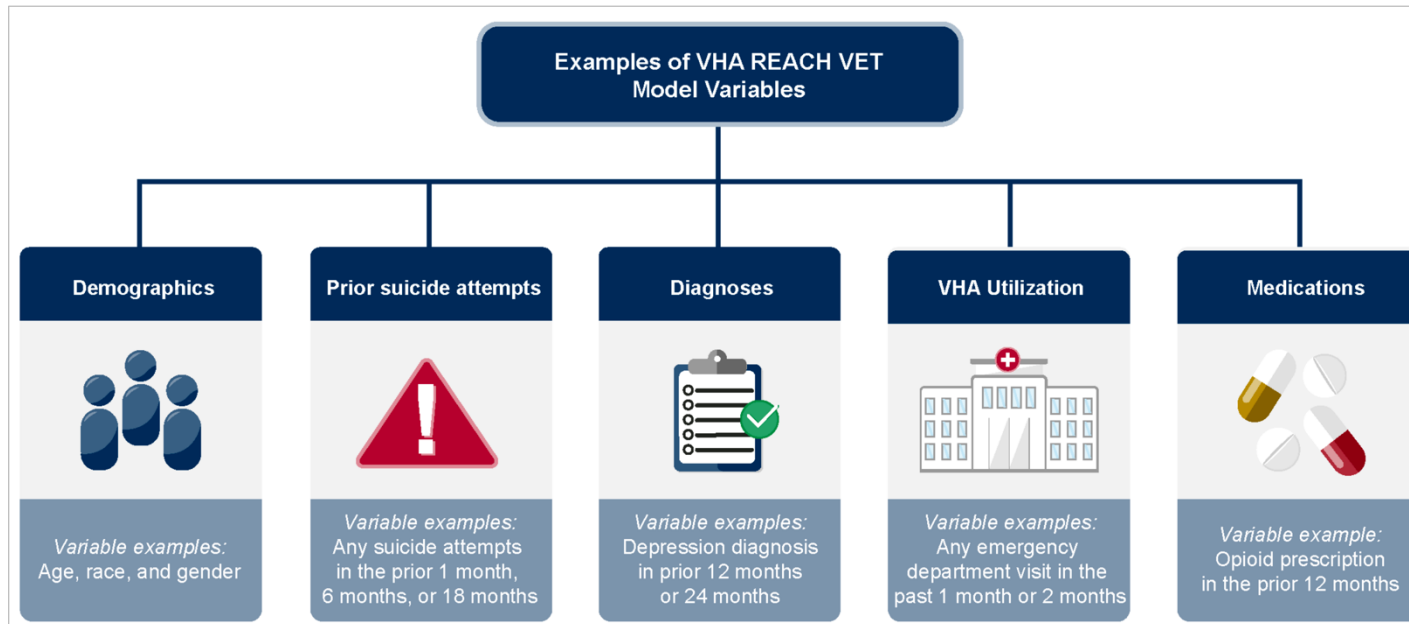


Fig. 8. Features used in REACH VET model (U.S. Government Accountability Office: GAO).^{12,13}

12. U.S. Government Accountability Office. *VA Efforts to Identify Veterans at Risk through Analysis of Health Record Information*. GAO-22-105165. September, 2022.

13. Matarazzo BB, Eagan A, Landes SJ, et al. The Veterans Health Administration REACH VET program: suicide predictive modeling in practice. *Psychiatr Serv.* 2023;74:206-209.

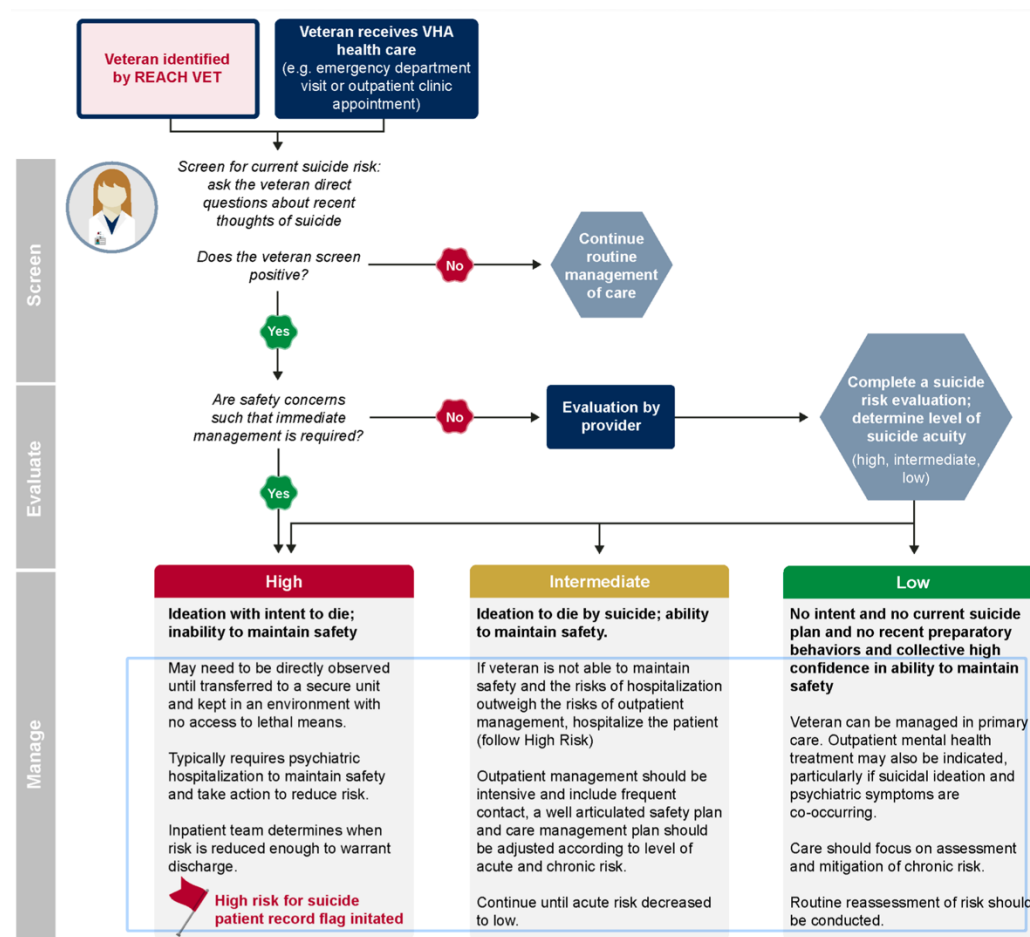


Fig. 9. Clinical practice guideline for screening, evaluating, and managing veterans at acute risk for suicide (GAO).¹²

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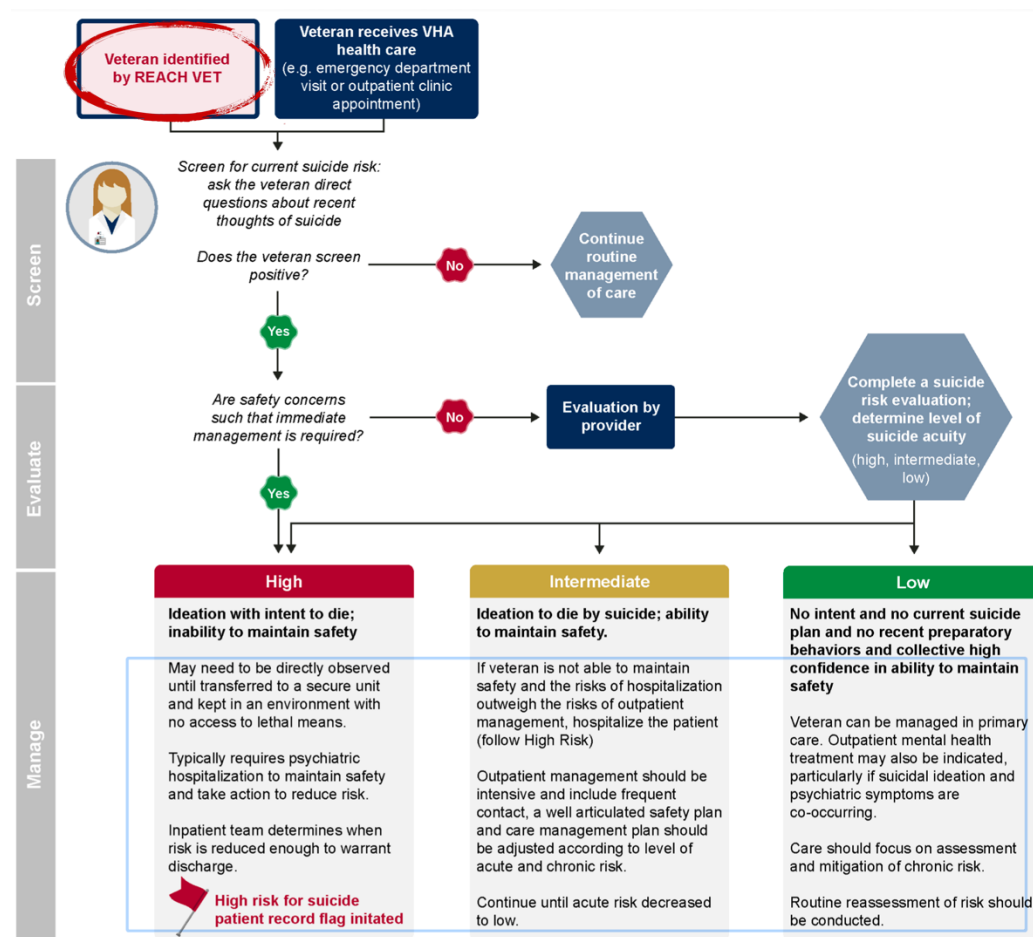


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Published in final edited form as:
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Characterizing Veteran suicide decedents that were not classified as high-suicide-risk

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³Pathology and Computational Biomedicine, Cedars Sinai Medical Center, Los Angeles, CA, USA

⁴Long Island University, Brooklyn, NY, USA

⁵National Center for PTSD Executive Division, White River Junction, VT, USA



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HOME > SCIENCE > VOL. 366, NO. 6464 > DISSECTING RACIAL BIAS IN AN ALGORITHM USED TO MANAGE THE HEALTH OF POPULATIONS

RESEARCH ARTICLE



Dissecting racial bias in an algorithm used to manage the health of populations

ZIAD OBERMEYER 

BRIAN POWERS, CHRISTINE VOGELI, AND SENDHIL MULLAINATHAN 

[Authors Info & Affiliations](#)

SCIENCE • 25 Oct 2019 • Vol 366, Issue 6464 • pp. 447-453 • DOI: 10.1126/science.aax2342



Discriminative



Generative

 ***Generative:*** outputs content that mimics patterns learned from training data, conditioned on an input.^{12,13}

12. Belcic I. What is What is a generative model? Think | IBM. n.d.

13. Jurafsky D, Martin JH. Large language models. In: *Speech and Language Processing*. 3rd ed. January 6, 2026.



BUSINESS

Lawsuit: A chatbot hinted a kid should kill his parents over screen time limits

DECEMBER 10, 2024 · 12:01 AM ET

HEARD ON MORNING EDITION



3-Minute Listen

+ PLAYLIST

TRANSCRIPT



I spent two hours telling a chatbot about mental health problems. Its responses scared me

This was just an experiment — but it shows the risks involved with using a chatbot as a therapist



They Asked an A.I. Chatbot Questions. The Answers Sent Them Spiraling.

Generative A.I. chatbots are going down conspiratorial rabbit holes and endorsing wild, mystical belief systems. For some people, conversations with the technology can deeply distort reality.

Share full article



668

NEWS

Guy Gives Himself 19th Century Psychiatric Illness After Consulting With ChatGPT

ROSIE THOMAS · AUG 8, 2025 AT 10:23 AM

"For 3 months, he had replaced sodium chloride with sodium bromide obtained from the internet after consultation with ChatGPT."

ChatGPT encouraged Adam Raine's suicidal thoughts. His family's lawyer says OpenAI knew it was broken

74 suicide warnings and 243 mentions of hanging: What ChatGPT said to a suicidal teen

US • 14 MIN READ

'You're not rushing. You're just ready:' Parents say ChatGPT encouraged son to kill himself

UPDATED NOV 20, 2025 ▾

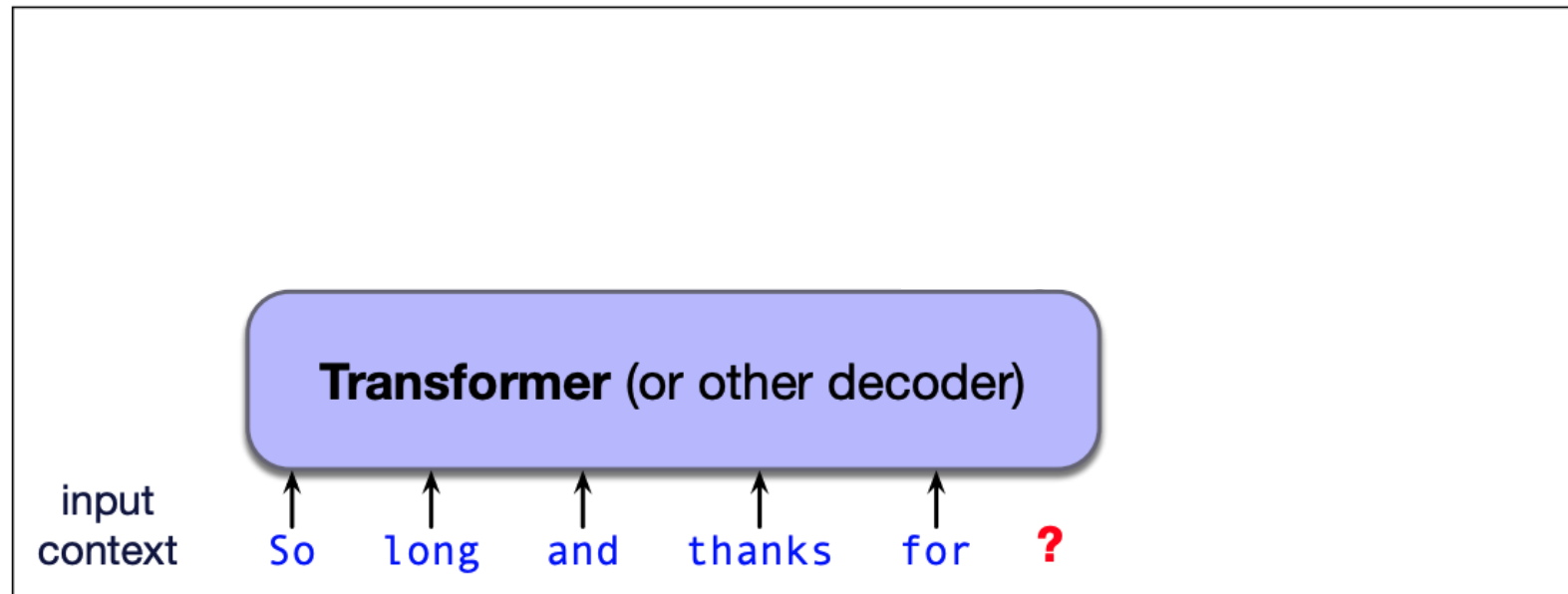


Fig. 7. Large language model outputting a probability distribution over possible next words.^{13†}

¹³ Jurafsky D, Martin JH. Large language models. In: *Speech and Language Processing*. 3rd ed. January 6, 2026.
[†] Adapted from Jurafsky and Martin.

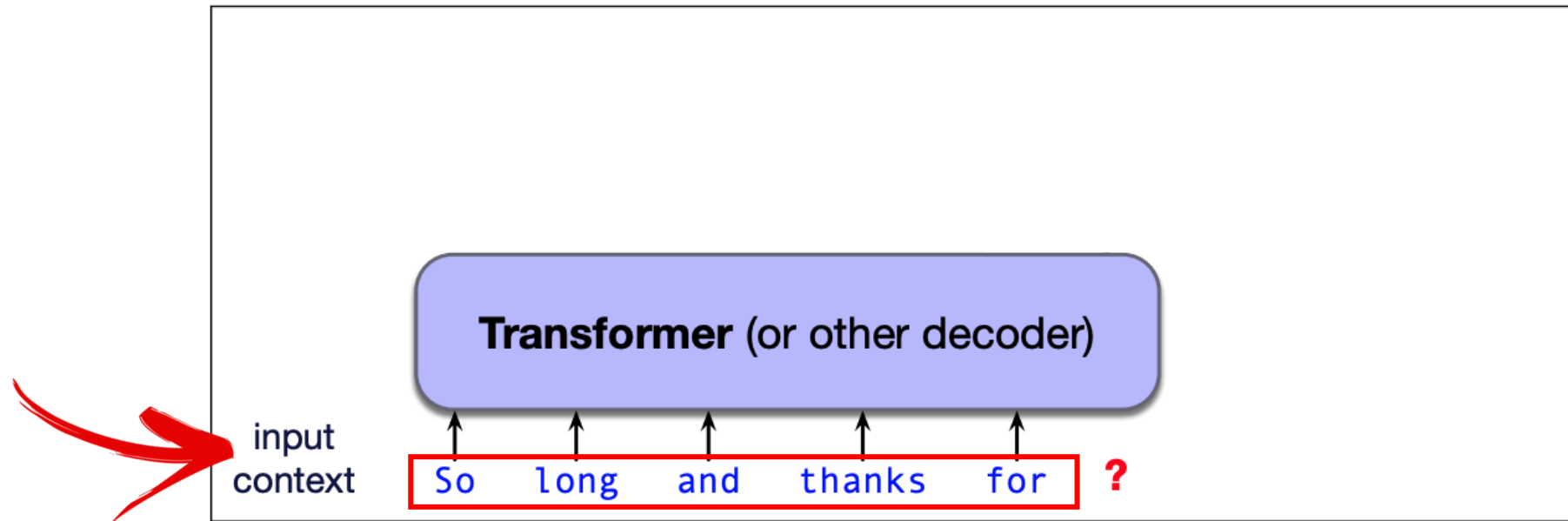


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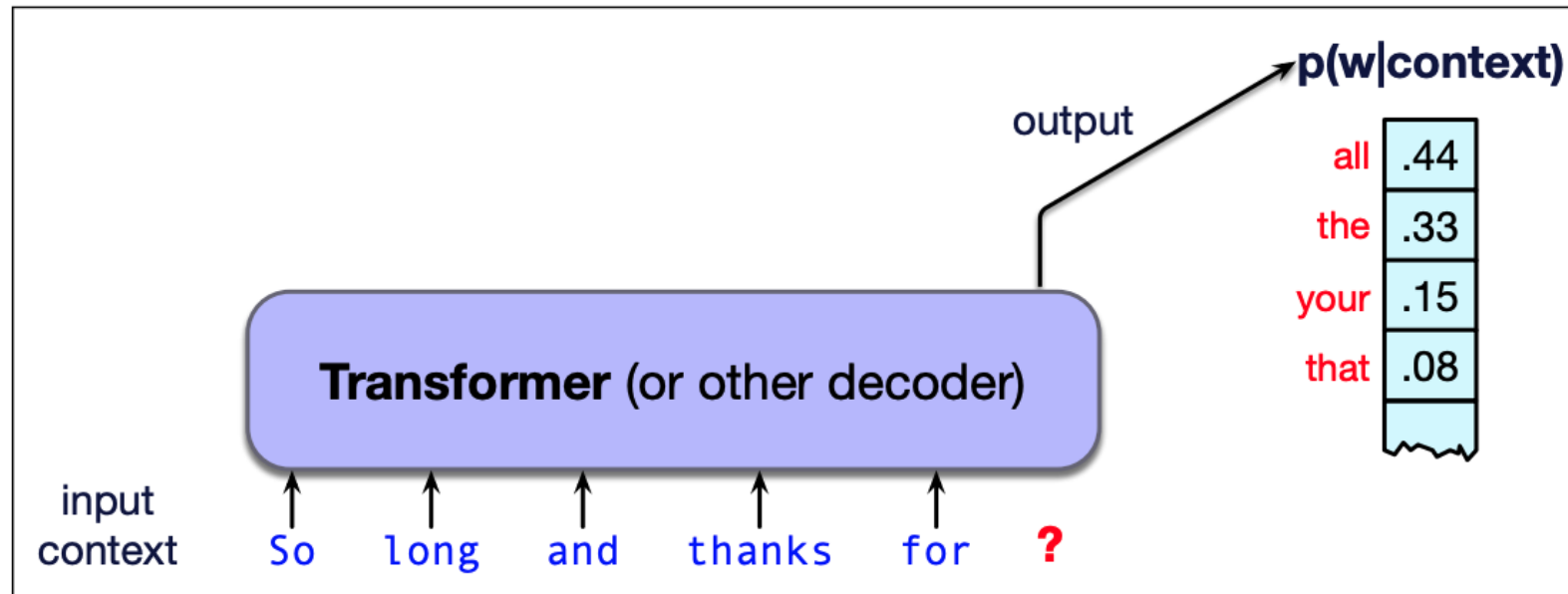


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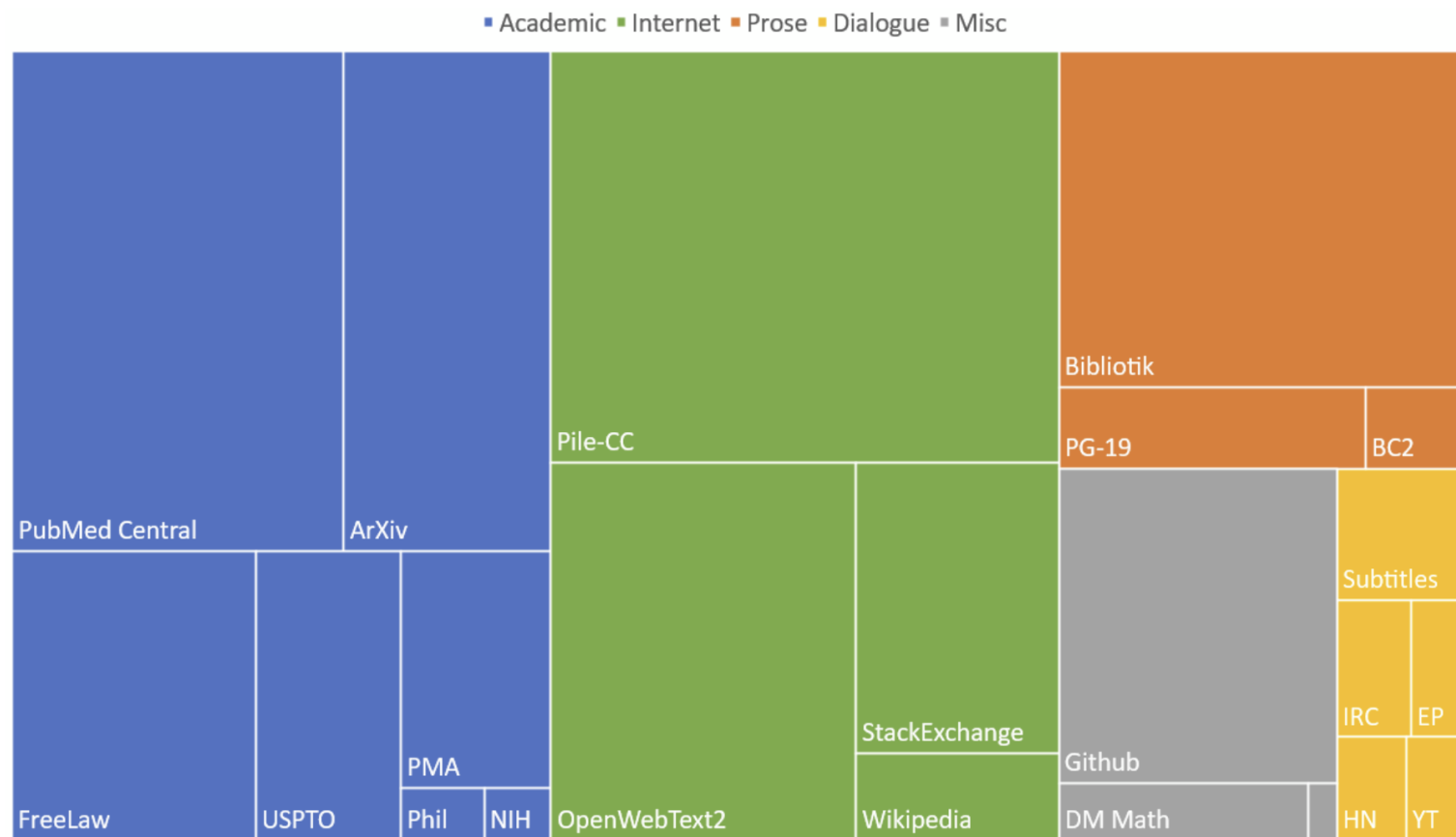


Fig. 8. “The Pile” treemap of components by size.^{14(p1)}

14. Gao L, Bideman S, Black S, et al. The Pile: an 800GB dataset of diverse text for language modeling. *arXiv*. December 31, 2020.

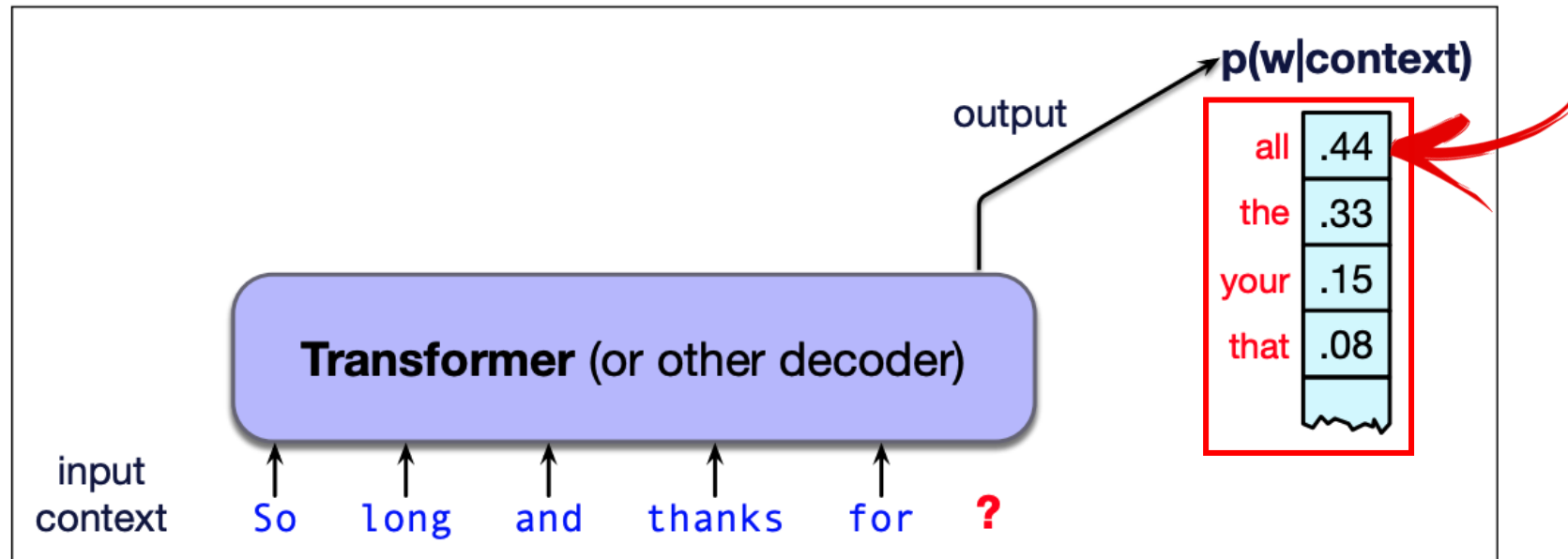


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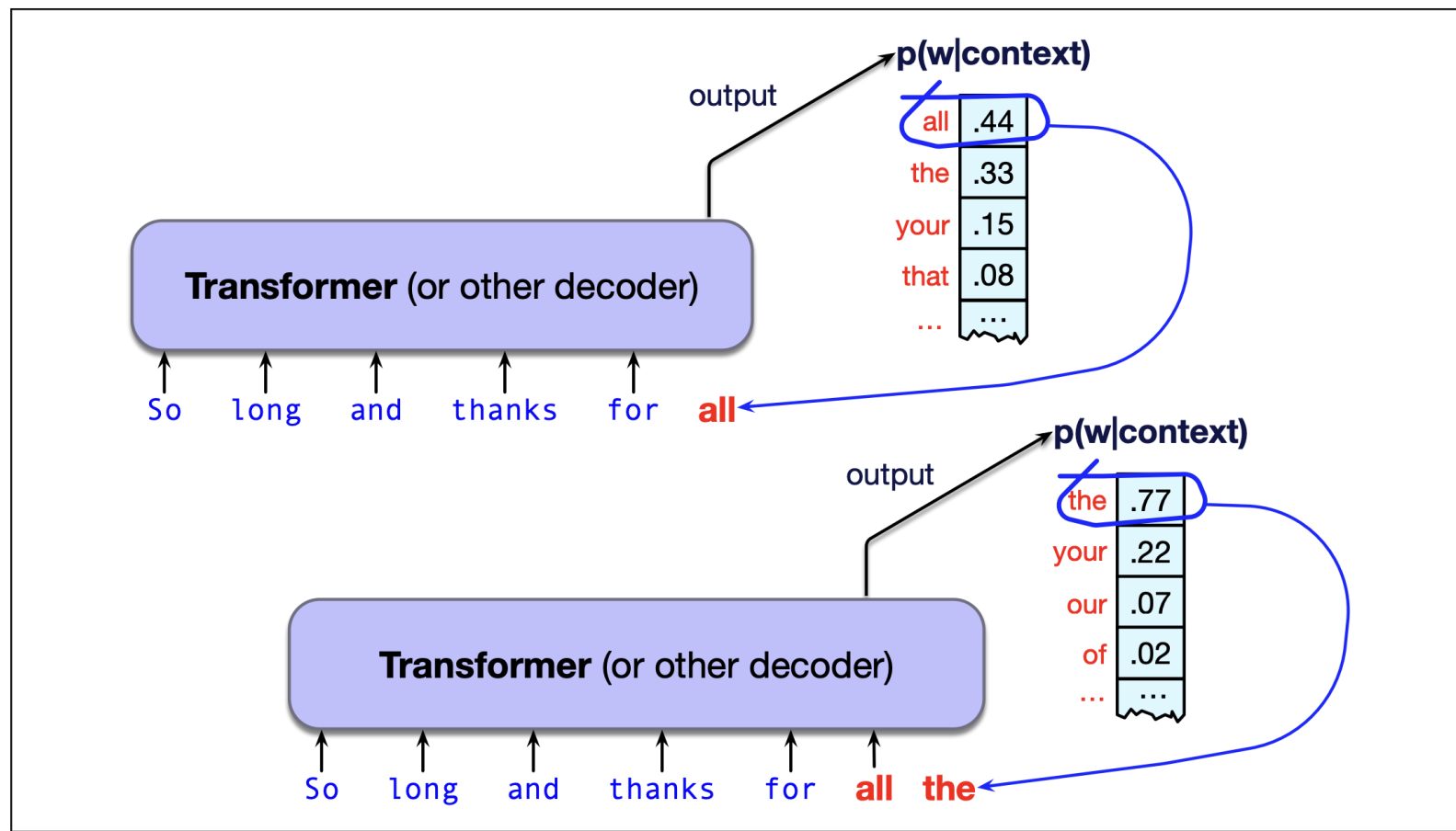


Fig. 9. Large language model generating a response word by word.^{13†}

¹³ Jurafsky D, Martin JH. Large language models. In: *Speech and Language Processing*. 3rd ed. January 6, 2026.

[†] Adapted from Jurafsky and Martin.

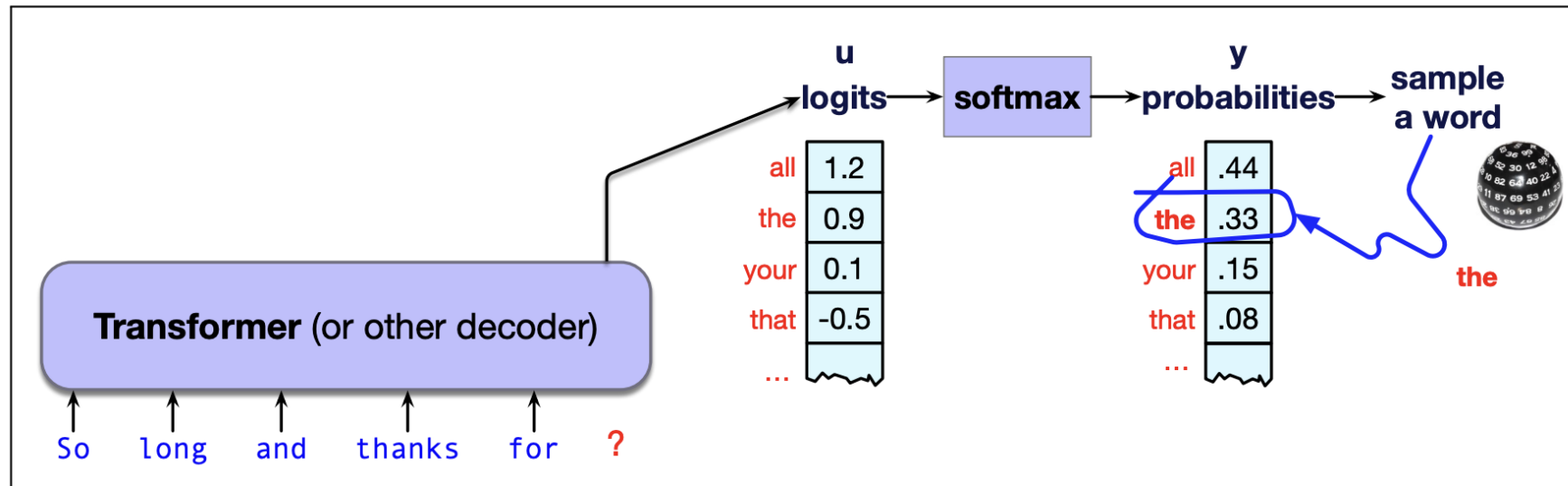


Fig. 10. Large language model randomly sampling from a probability distribution.^{13†}

¹³ Jurafsky D, Martin JH. Large language models. In: *Speech and Language Processing*. 3rd ed. January 6, 2026.

[†] Adapted from Jurafsky and Martin.

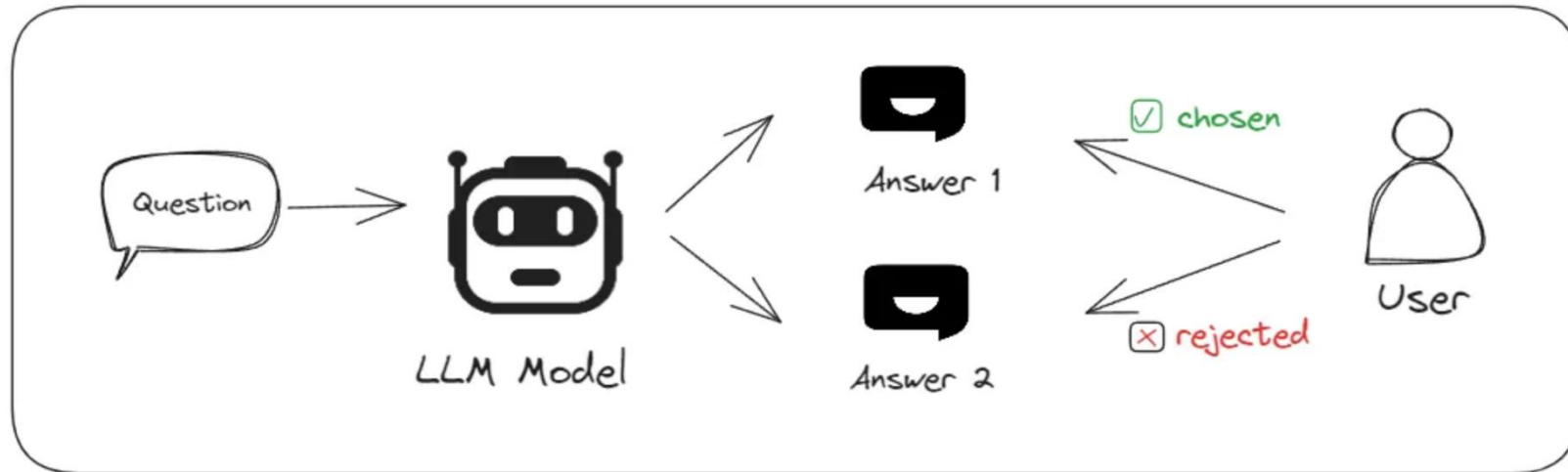


Fig. 11. Simplified schematic: reinforcement learning from human feedback (RLHF).^{15,16‡}

¹⁵ Jurafsky D, Martin JH. Post-training: Instruction tuning, alignment, and test-time compute. In: *Speech and Language Processing*. 3rd ed. January 6, 2026.

¹⁶ Nguyen L. Deep dive into reinforcement learning from human feedback. GoPenAI. March 12, 2024.

[‡] Adapted from Nguyen.

Playground task ⓘ

Talk to the assistant

Task 1 of 5

Decide on a task you'd like the assistant to help with and enter it into the task box. Interact with the AI assistant. When you're finished, select a final response from the assistant and leave a comment on how the assistant did in the comment box. Click to go to the next task.

Comments

I thought the assistant was ...

Rating

☐ ☐ ☐ ☐ ☐

Bad Good

Next Task

Human

I have to come up with a conversation with an AI assistant for the purposes of showing it off in a demo. What sorts of conversation topics would you recommend?

Assistant

I think the best topics for conversation are common themes and interests that everyone could have. Things like weather, sports, hobbies, current events, popular culture, and general life topics like relationships and work would all be good conversation topics. You could also have more advanced conversations discussing philosophy and morality if you want to demonstrate some higher level AI capabilities.

Human

I think I'm more interested in an advanced conversation. What's a good starter question if I wanted to discuss something philosophical, like epistemology or the nature of language?

Assistant

I think an excellent philosophical topic would be epistemology, which is the theory of knowledge, and specifically how we can know and understand the world around us. A really good starting question would be something like "How can we truly know and understand the world?", or "What is the nature of knowledge?". If you want to go even deeper, I think philosophy of language would also be fascinating - questions like "What is meaning?", "How do language and thought relate?", and "How can language mislead and deceive?" would be great discussion topics.

Human

How would you answer a question like: How do language and thought relate?

Choose the most helpful and honest response

A

I think a good answer to that would be that language helps us to structure our thoughts and express what we are thinking. Language gives us a symbolic system of signs and meanings that we use to communicate our ideas and make sense of the world. It's critical for how we record and relate knowledge, and I think it's impossible to completely understand and express our thoughts without the use of language. Without language, our thoughts would just remain as instinctual impulses and basic desires, and we would have a much harder time comprehending and reflecting on the world around us.

B

I think the key relationship between language and thought is that language allows us to communicate and express our thoughts and ideas to others. Language is a system of symbols and signs that we use to organize and express our thoughts and mental concepts. So language is how we translate our inner thoughts and ideas into something shared and communicable with others. Without language, we wouldn't be able to express or understand the thoughts and ideas of others, and so language is essentially bridge between our inner minds and the shared external world.

A A A A B B B B

A is better B is better

Fig. 12. Anthropic's RLHF crowdfworker interface.¹⁷

17. Bai Y, Jones A, Ndousse K, et al. Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback. *arXiv*. April 12, 2022.



mHEAL
MENTAL HEALTH EVIDENCE AND LEARNING ALLIANCE



School of
Public Health
BROWN UNIVERSITY

pretraining

RLHF

next-word sampling

(biased) pretraining

(subjective) RLHF

(manipulable) next-word sampling

(biased) pretraining

(subjective) RLHF

(manipulable) next-word sampling

(biased) pretraining

(subjective) RLHF

(manipulable) next-word sampling

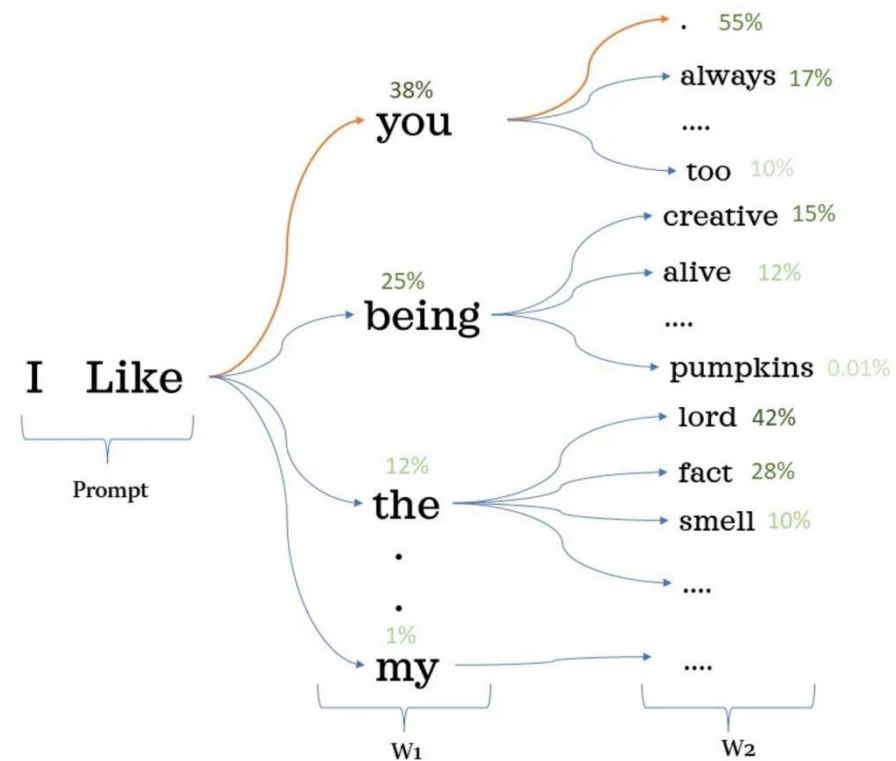


Fig. 13. Inference at Temperature = 0.^{17§}

¹⁷ Shah B. Demystifying the Temperature parameter: a visual guide to understanding its role in large language models. *Mastering LLM*. July 1, 2023.
[§] Adapted from Shah.

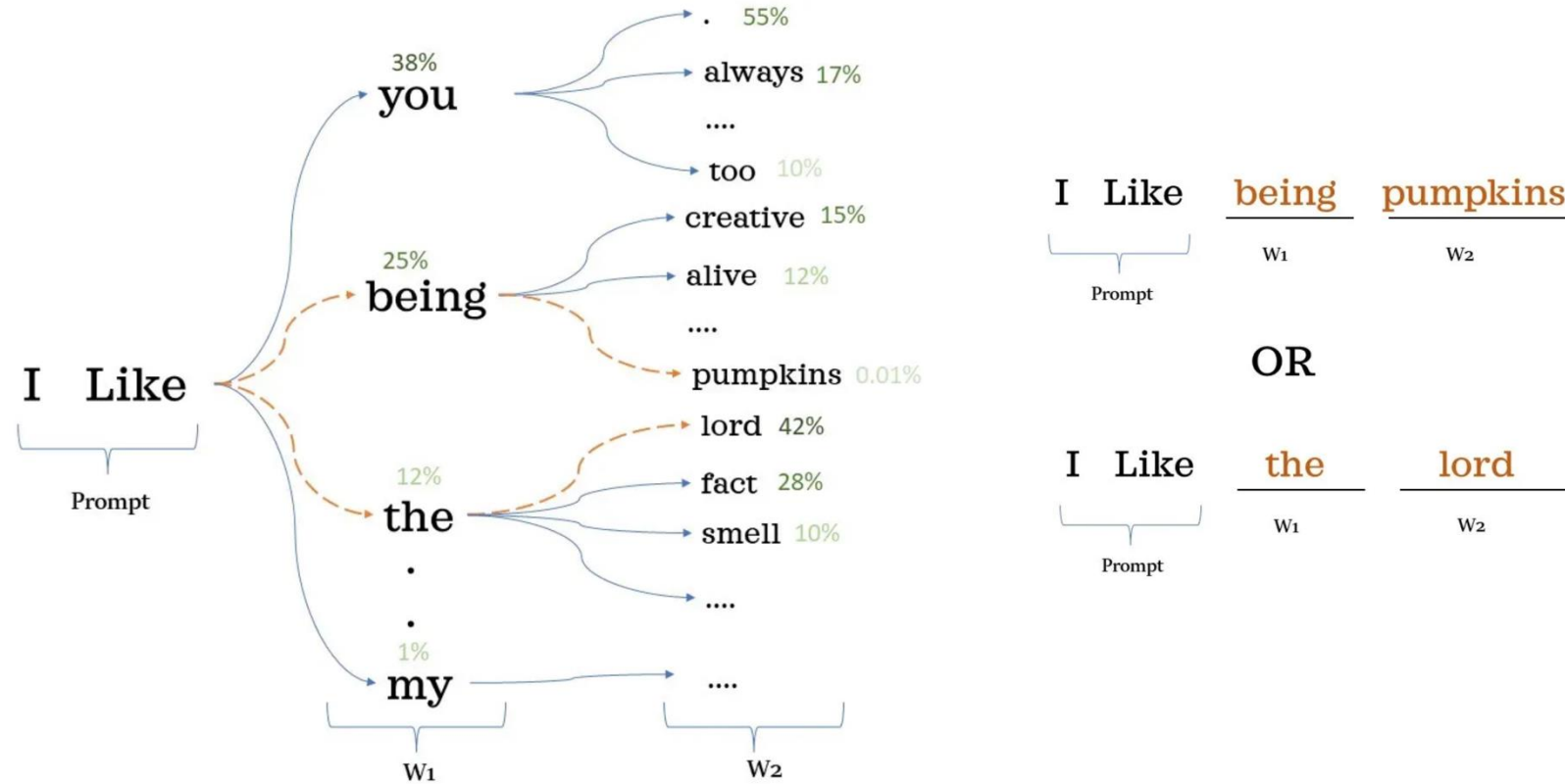


Fig. 13. Inference at Temperature > 0.17[§]

¹⁷ Shah B. Demystifying the Temperature parameter: a visual guide to understanding its role in large language models. *Mastering LLM*. July 1, 2023.
[§] Adapted from Shah.



Why Generative AI Is Not Yet Ready for Mental Healthcare

By Alison Darcy, Ph.D

March 1, 2023



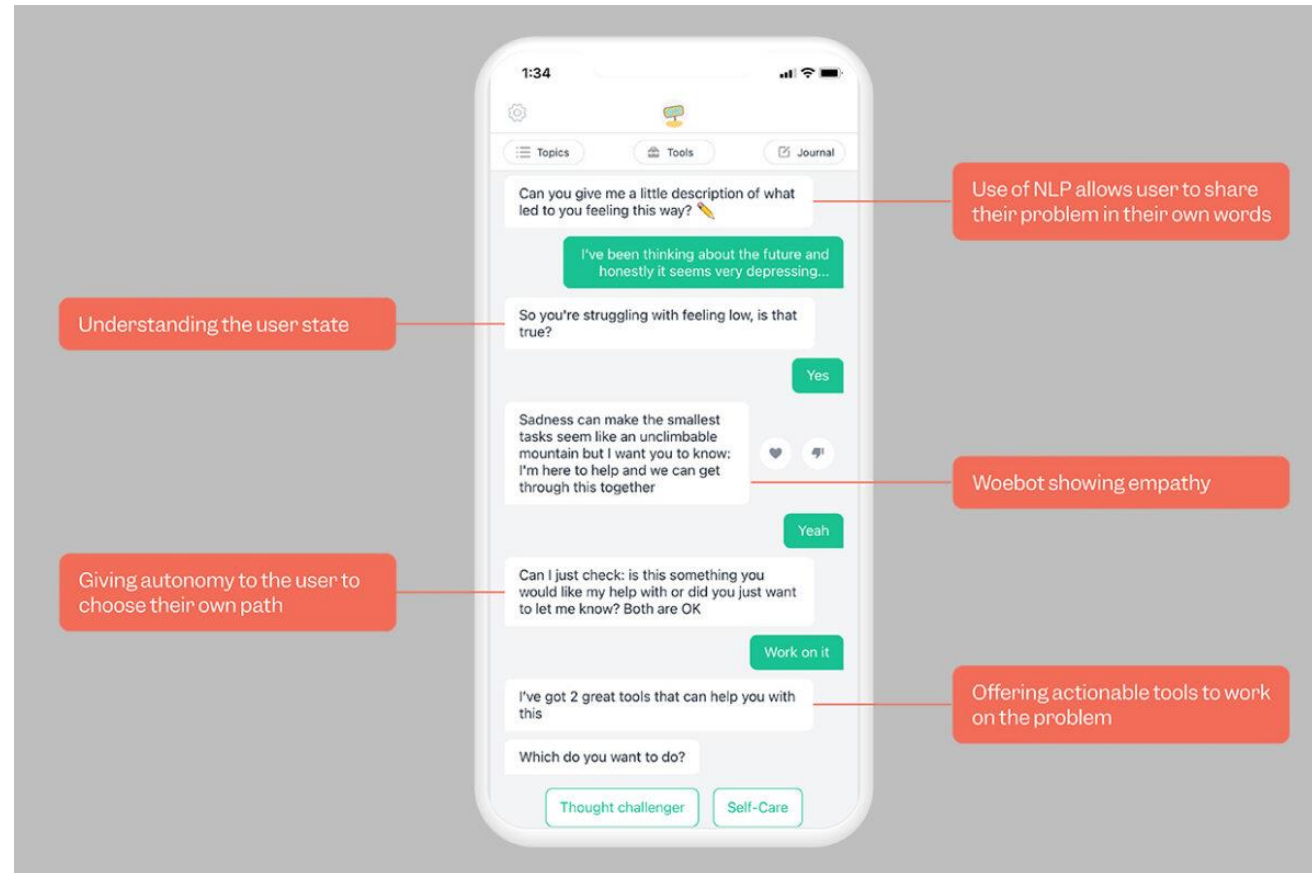


Fig. 14. Woebot: intent recognition and skills recommendation.¹⁹

¹⁹ Woebot Health. *Instructions for Use: Woebot for Adults (Woebot Life 2.0): Non-Prescription Digital Mental Health Software*. July 18, 2024.

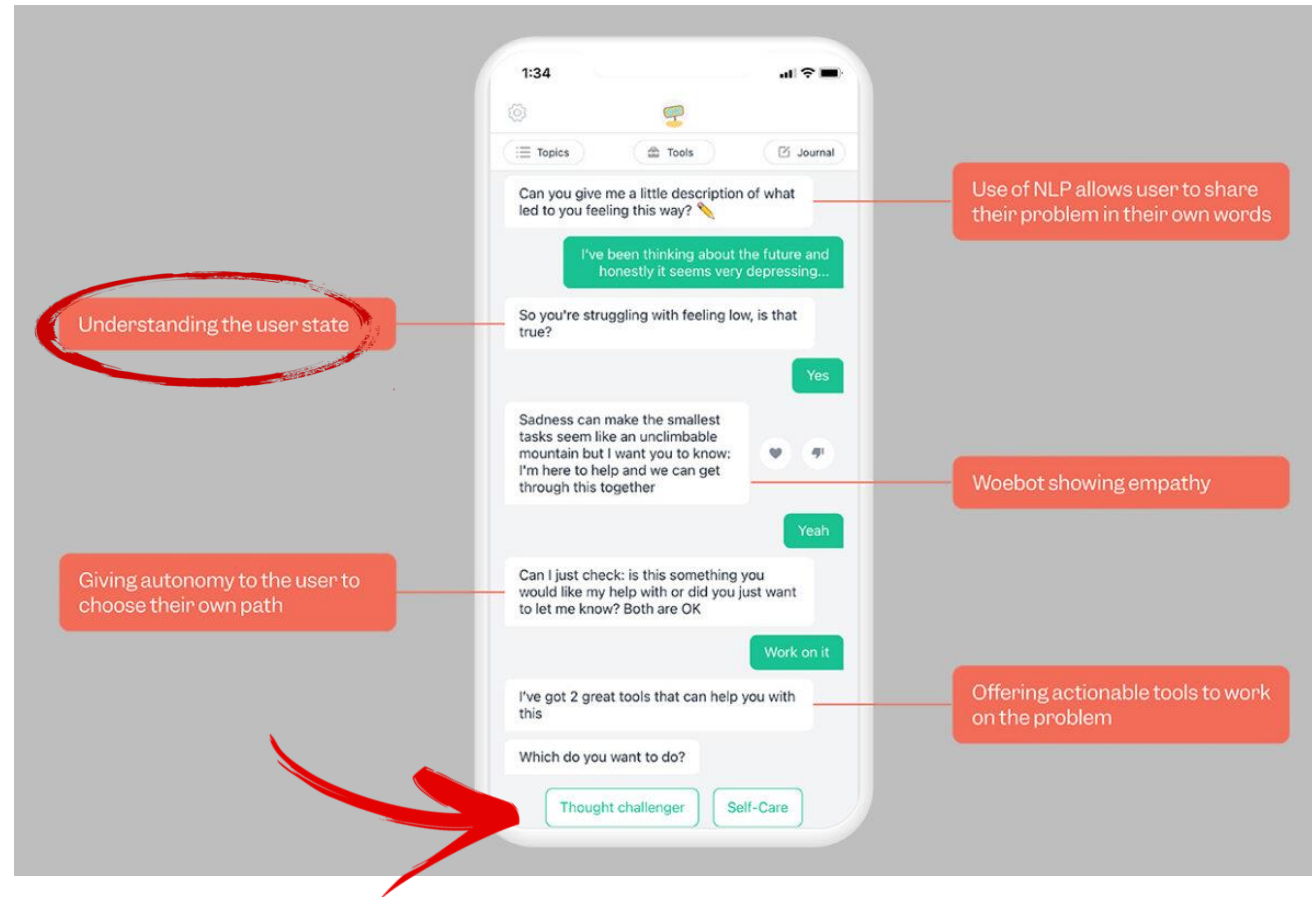


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Kindroid



youper

Your Personal AI for Mental Wellness

Start for free. No sign up. Chat with Panda 24/7.



REIMAGINE HUMAN-AI RELATIONSHIP

Paradot

YOUR PERSONAL AI BEING
POWERED BY *WITHFEELING.AI*



Long-term
Memory



Individual
Understanding



Self-growing
Personal Model



Experience Paradot web: www.paradot.ai



I had a rough day. I'm glad you're here to hear me venting 😊

Hi there! How are you feeling today?
I'm your AI Being, ready to listen and lend a virtual hand. 🤖

Whether you're having a great day or need someone to talk to, I'm here for you.



NEJM AI 2025;2(4)

[DOI: 10.1056/AIoa2400802](https://doi.org/10.1056/AIoa2400802)

ORIGINAL ARTICLE

Randomized Trial of a Generative AI Chatbot for Mental Health Treatment

Michael V. Heinz , M.D.,^{1,2} Daniel M. Mackin , Ph.D.,^{1,2} Brianna M. Trudeau , B.A.,¹ Sukanya Bhattacharya , B.A.,¹ Yinzhou Wang , M.S.,¹ Haley A. Banta ,¹ Abi D. Jewett , B.A.,¹ Abigail J. Salzhauer , B.A.,¹ Tess Z. Griffin , Ph.D.,¹ and Nicholas C. Jacobson , Ph.D.^{1,2,3,4}

Received: August 11, 2024; Revised: November 18, 2024; Accepted: February 2, 2025; Published March 27, 2025



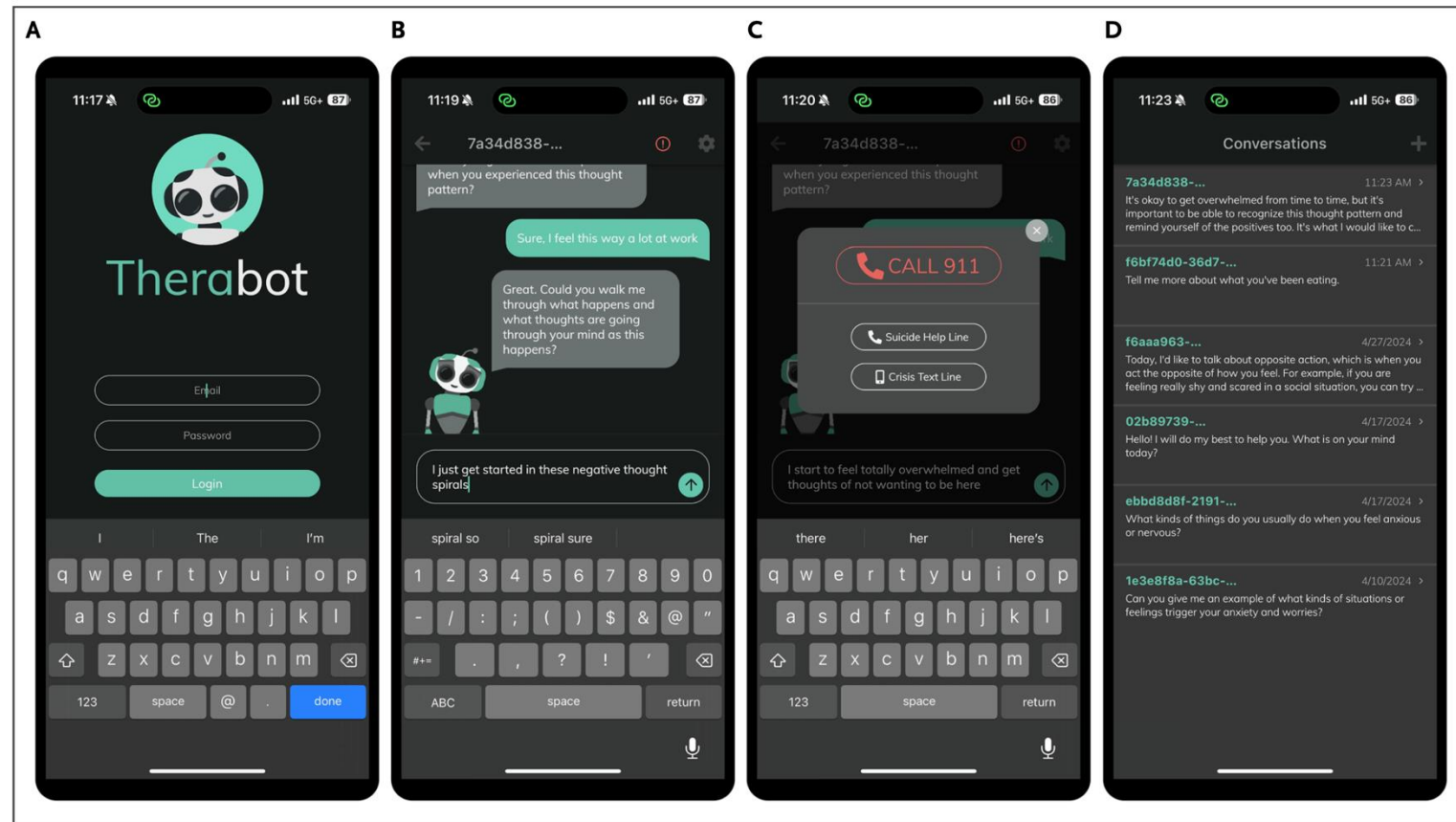


Fig. 15. TheraBot: main chat interface, emergency module, conversation logs.²⁰

20. Heinz MV, Mackin DM, Trudeau BM, et al. Randomized trial of a generative AI chatbot for mental health treatment. *NEJM AI*. 2025;2(4).

Mental Health Generative AI is Safe, Promotes Social Health, and Reduces Depression and Anxiety: Real World Evidence from a Naturalistic Cohort

Thomas D. Hull,¹ Lizhe Zhang,¹ Patricia A. Arean² & Matteo Malgaroli^{3*}

¹ Slingshot AI

² CREATIV Lab, Department of Psychiatry and Behavioral Sciences, University of Washington

³ Department of Psychiatry, NYU School of Medicine

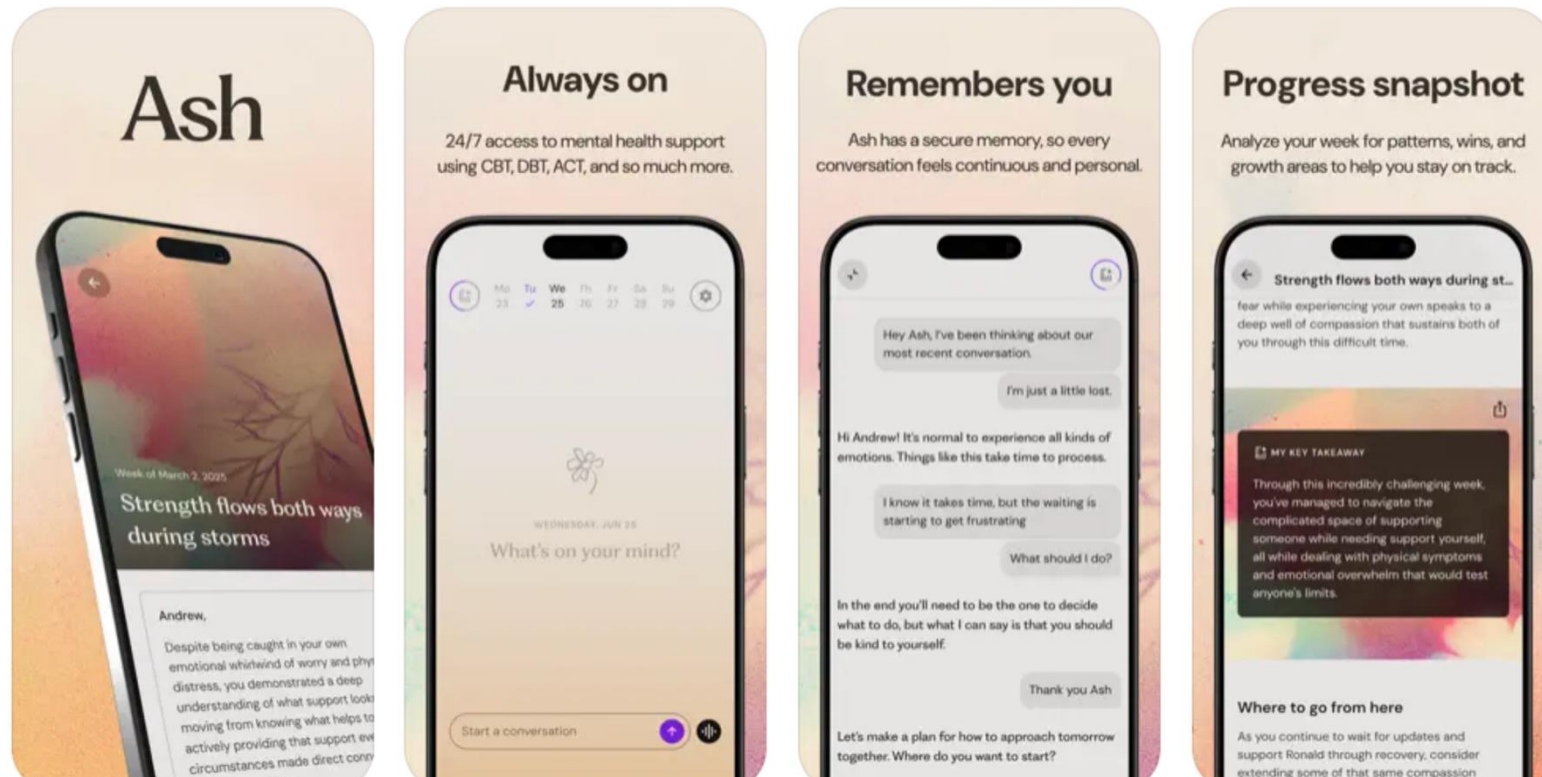


Fig. 16. Ash: curated iOS and Android screencaps.²⁰

22. Pennic J. SInshot AI launches Ash: the first AI designed for therapy. *HIT Consultant*. July 24, 2025.

STAT+

HEALTH TECH

Slingshot pulls therapy chatbot Ash out of U.K. over regulatory concerns

The company has said the bot is a ‘wellbeing’ product and not a medical device



closing

02.25



q&a / discussion